Vouch — Business Plan

A Peer Tutoring Marketplace with a Verifiable Reputation Record

CS 4800-02 · Fall 2026 · Course Project (Option 04, Track A)

Working product name; substitute your team's chosen name throughout.

1. Executive Summary

1.1 Business Concept

Vouch is a campus-scoped marketplace that connects students who need academic help with peer tutors who have already succeeded in the same courses. It differs from general tutoring platforms in two ways. First, matching is course-specific and availability-aware rather than subject-general — a student searching for help does not get a list of "math tutors," they get ranked peers who have completed the exact course, with the exact professor where possible, and whose free hours actually intersect their own.

Second, and more central to the product, the reputation record is tamper-evident. Reviews are written into an append-only log in which each entry cryptographically incorporates the hash of the entry before it. Altering or deleting a past review breaks the chain at that point and every verification afterward fails. Reviews can only be created against an attested session — a booking both parties confirmed took place — which closes the loop that lets fake reviews accumulate on open platforms.

The pitch to a campus is therefore not "we have tutors." It is "we have a reputation record you can audit."

1.2 Target Market

The initial market is a single large public commuter university, modeled on Cal Poly Pomona: roughly 30,000 students, heavily represented in engineering, computer science, and business, with a high proportion of transfer and first-generation students and a documented set of high-DFW-rate gateway courses (calculus sequences, general chemistry, physics, data structures).

Expansion targets are institutions with the same shape — large CSU and regional public campuses where the official tutoring center is oversubscribed and the informal market already exists in Discord servers and group chats but is unstructured, unaccountable, and invisible to the institution.

The buyer and the user are not always the same. Students are the users. The paying customer in the institutional model is the department, learning center, or student success office that wants supplemental instruction capacity it can report on.

1.3 Proposed Solution

A web application with four subsystems:

1. **Profiles and availability.** Tutors declare completed coursework, grade earned, teaching modality, and a weekly availability model. Students declare current enrollment and learning preferences.
2. **Matching engine.** A deterministic, weighted scoring function ranks tutors against a structured help request. A language model handles only the intake step — turning a free-text request ("lost on dynamic programming, midterm Thursday") into structured fields — and the explanation step, generating the human-readable reason a given tutor was ranked where they were. It does not compute the ranking.
3. **Booking and attestation.** Requests become bookings; bookings become sessions; sessions require confirmation from both parties before a review can be written. No-show policy and cancellation windows are enforced by the system, not by convention.
4. **Tamper-evident reputation ledger.** Reviews and attestations are appended to a hash-linked log. A verification routine walks the chain and reports the first broken link. The administrative interface includes a deliberate tamper action for demonstration and audit-testing purposes.

Post-session, generated summaries capture concepts covered and unresolved gaps, and those gaps feed back into the student's profile so subsequent matches improve. Summaries are advisory content and are stored outside the ledger;

nothing that a model generates participates in hash computation or verification.

1.4 Team

The project is delivered by two sub-teams across three four-week sprints.

Sub-team	Ownership
Team A — Intelligence	Structured intake, skill taxonomy, matching engine, ranking explanations, session summaries, match-quality metrics
Team B — Reputation & Booking	Booking flow, session attestation, review capture and anchoring, hash chain and verification, dispute records, no-show policy

Role	Member	Responsibility
Project Lead / Scrum Master	*TBD*	Sprint planning, backlog, stakeholder demo
Backend / Data	*TBD*	Schema, API, ledger implementation
Matching & ML	*TBD*	Scoring function, embeddings, evaluation harness
Frontend / HCI	*TBD*	Interface, accessibility, usability testing
QA & Documentation	*TBD*	Test plan, verification tests, SRS and architecture docs

2. Product Overview

2.1 Core Features

Sprint 1 (weeks 1–4) — Foundation

- Student and tutor profiles with verified course history
- Weekly availability model with timezone and recurring-block support
- Help request creation with structured fields
- Manual matching, so the workflow is exercised end-to-end before the engine exists

Sprint 2 (weeks 5–8) — Engine and Ledger

- Weighted matching engine with ranked results and explanations
- Booking lifecycle: request → accept → scheduled → completed
- Dual-party session attestation
- Review capture, anchored to an attested session
- Hash-chained append-only review log with a verification endpoint

Sprint 3 (weeks 9–12) — Integrity and Measurement

- Dispute filing, evidence attachment, and resolution records
- No-show policy with automatic penalty application
- Match quality metrics: acceptance rate, completion rate, repeat-booking rate, post-session confidence delta
- Administrative audit view with tamper demonstration

Matching score. Ranking is a transparent linear combination, tunable and defensible:

```
score = 0.40 · topic_match
      + 0.30 · availability_overlap
      + 0.15 · learning_style_fit
      + 0.15 · normalized_tutor_rating
      − 0.10 · current_load_penalty
```

`topic_match` is cosine similarity between the request's embedded topic text and the tutor's declared skill vector, mapped onto a canonical course taxonomy. `availability_overlap` is the fraction of the student's requested window

covered by the tutor's free blocks. The load penalty exists so that a small number of highly rated tutors do not absorb all demand and degrade response time.

2.2 Technology Stack

Layer	Choice	Rationale
Frontend	React + TypeScript, Tailwind	Component reuse, type safety across a five-person team
Backend	Node.js + Express (or FastAPI)	Fast iteration, broad team familiarity
Database	PostgreSQL	Relational integrity for bookings; append-only ledger table with a database-level rule preventing UPDATE and DELETE
Ledger	SHA-256 hash chaining in application code	Each record stores `prev_hash`; verification recomputes the chain
AI services	Hosted LLM API for structured intake, ranking explanations, and session summaries; embedding model for topic-to-taxonomy mapping	No training, no GPU, no labeled dataset required
Auth	Campus SSO in production; email + role-based access for the prototype	Institutional identity is what makes reviews meaningful
Testing	Jest / PyTest, plus a dedicated ledger-integrity suite	Verification correctness is the highest-value test target
CI / Hosting	GitHub Actions, containerized deploy	Sprint demo reliability

Model outputs shown in demonstrations are cached, so no live API call sits on the critical path of a graded presentation.

---

3. Market Analysis

3.1 Industry Overview

The global private tutoring market is large and growing, though published estimates vary widely by research firm — a range worth stating plainly rather than citing a single figure as fact. The Business Research Company places the market at approximately $131 billion in 2025, growing to roughly $143 billion in 2026 at about a 9.1% CAGR, and projects roughly $209 billion by 2030. IMARC Group estimates approximately $134 billion for 2025; Fortune Business Insights estimates roughly $67 billion for the same year using a narrower definition. The consistent signals across all of them are high single-digit to low double-digit growth, rapid migration from in-person to online delivery, and AI-driven personalization as the dominant product trend.

In the United States, roughly one in five students receives some form of private tutoring, and the large majority of tutoring programs now offer online delivery. Peer tutoring specifically sits in a gap: campus learning centers provide it free but with limited hours, limited course coverage, and long waits, while commercial platforms charge $40–$80 per hour, which is out of reach for the students who most need help.

3.2 Target Market & Competitive Analysis

Serviceable market at one campus. With approximately 30,000 students, assume 20% seek paid academic help in a given term (6,000 students), at an average of 6 sessions per term at $20 per session. That is roughly $720,000 in per-term transaction volume at a single institution. This is an order-of-magnitude estimate built from stated assumptions, not a forecast; each assumption should be tested against campus survey data before it is relied on.

Competitor	Model	Weakness Vouch addresses
Wyzant	National tutor marketplace, $40–$80/hr	Price; tutors have no course-specific or professor-specific context
Varsity Tutors	Subscription + on-demand	Not peer-based; no institutional integration
Knack	Campus peer tutoring, institution-funded	Closest competitor; reputation is platform-asserted rather than independently verifiable
Campus tutoring center	Free, staffed, limited hours	Capacity-constrained; narrow course coverage; no scheduling flexibility
Discord / GroupMe / word of mouth	Free, informal	No accountability, no record, no discovery beyond one's existing network

The realistic incumbent is the last row. Most peer tutoring already happens informally; the product does not need to create demand, it needs to make an existing behavior discoverable and accountable.

Defensibility. The hash-chained record is straightforward to replicate technically. The durable advantage is the accumulated reputation graph within an institution and, if achieved, integration with campus identity and course enrollment data — which is an institutional relationship rather than a piece of code.

---

4. Marketing and Sales Strategy

4.1 Marketing Channels

- **Course-level targeting.** Concentrate launch effort on the ten highest-DFW-rate courses rather than the campus as a whole. Density inside one course is worth more than thin coverage across two hundred.
- **Faculty and advisor referral.** Instructors of gateway courses are a credible, low-cost distribution channel and are motivated by their own pass rates.
- **Student organizations.** Discipline-specific clubs (SWE, ACM, NSBE, SHPE, business fraternities) already contain the supply side.
- **Existing informal channels.** Presence in the Discord servers and group chats where the behavior already occurs, positioned as infrastructure rather than as a replacement.
- **Physical presence.** Tabling during weeks 4–6 and 10–12, when midterm and final pressure peaks.

4.2 Customer Acquisition

Supply is seeded first. A marketplace with tutors and no students is recoverable; the reverse is not. Recruit 40–60 tutors in the top-ten course list before opening to students, using guaranteed-hours incentives and transcript-verified badges.

Acquisition follows a two-sided sequence:

1. **Supply seeding (weeks 1–4):** direct recruitment through faculty referral and club partnerships.
2. **Demand launch (weeks 5–8):** targeted at students in the same courses, timed to the first midterm.
3. **Referral loop:** both parties receive session credit for a referral that completes a first session.

Cost per acquired student in the campus model is near zero in cash and non-trivial in labor — the honest framing is that early acquisition is founder time, not marketing spend.

4.3 Growth Strategy & Monetization

Monetization options, in order of realism for a campus product:

1. **Transaction fee (primary).** A 12% platform fee on peer-to-peer sessions. At a $20 session that is $2.40 in revenue. At 400 sessions per month, monthly revenue is 400 × $2.40 = $960; at 2,000 sessions per month it is $4,800. Against roughly $200–$400 per month in hosting and API costs at that volume, the model reaches contribution-positive somewhere in the low hundreds of monthly sessions — the sensitivity is almost entirely to session volume, not to price.

2. **Institutional license (highest value).** An annual per-campus fee, in the $15,000–$40,000 range, sold to a student success office that wants supplemental instruction capacity plus reporting. One institutional contract is worth more than a full year of transaction fees at realistic student volume, which means the long-term business is B2B even though the user experience is B2C.
3. **Tutor subscription (secondary).** A monthly fee for profile promotion and analytics. Modest revenue; introduces a ranking-integrity conflict that should be handled explicitly — paid placement must be labeled and must not enter the match score.

Growth path. Establish depth in ten courses at one campus, expand course coverage to the full STEM core, then expand to a second and third campus in the same system where the course catalog and student profile are close enough that the taxonomy transfers with minimal rework. Multi-campus expansion is where the reputation ledger becomes a genuine asset: a portable, verifiable record that follows a tutor across institutions.

Key risks.

- *Cold start.* Mitigated by course-level concentration rather than campus-wide launch.
- *Disintermediation.* Once matched, both parties can transact off-platform. Mitigated by making the reputation record valuable enough that tutors want sessions logged, not by attempting to prevent contact.
- *Institutional policy.* Some campuses restrict paid tutoring by enrolled students in courses they have taken. This must be verified with the relevant office before any real launch and is a genuine unknown, not a formality.
- *Academic integrity.* Tutoring adjacent to coursework can shade into completing coursework. Policy, session summaries, and dispute records exist partly to make that boundary auditable.

Market figures in section 3.1 are drawn from published industry reports as of 2026 and should be re-verified before submission. Financial projections throughout are illustrative models built from stated assumptions rather than forecasts.